Experiment No 5

Course Outcome: CO3 **Blooms Level: L5**

Aim: To evaluate the performance of different machine learning models using standard evaluation metrics.

Abstract: In real-world data analysis, selecting appropriate machine learning models and accurately evaluating their performance is a critical challenge, as different learning paradigms require different validation strategies. This experiment addresses the problem of developing and systematically evaluating supervised and unsupervised learning models using the Wine Dataset. The task involves predicting a continuous variable using regression techniques, classifying wine samples into predefined categories using classification algorithms with ROC–AUC analysis, and identifying inherent group structures using clustering methods. The performance of each model is assessed using relevant evaluation metrics such as MAE, MSE, RMSE, R^2 score, Accuracy, Precision, Recall, F1-score, Confusion Matrix, ROC–AUC curve, Silhouette Score, Davies–Bouldin Index, Calinski–Harabasz Index, and Adjusted Rand Index. The experiment aims to analyze and compare the effectiveness of different evaluation measures across regression, classification, and clustering tasks to ensure reliable and meaningful model validation.

Theory:

Regression: Regression predicts a continuous value.

Evaluation Metrics:

- MAE (Mean Absolute Error) – Average absolute difference between actual and predicted values.
- MSE (Mean Squared Error) – Average squared difference.
- RMSE (Root Mean Squared Error) – Square root of MSE.
- R^2 Score – Proportion of variance explained by the model.

Classification: Classification predicts categorical labels.

Evaluation Metrics:

- Accuracy – Ratio of correct predictions to total predictions.
- Precision – Correct positive predictions / Total predicted positives.
- Recall – Correct positive predictions / Total actual positives.
- F1 Score – Harmonic mean of precision and recall.
- Confusion Matrix – Matrix showing actual vs predicted values.
- ROC Curve (Receiver Operating Characteristic Curve) – Plots TPR vs FPR.
- ROC–AUC Score – Area under ROC curve (measures classification separability).

Clustering: Clustering groups similar data points without labels.

Evaluation Metrics:

- Silhouette Score – Measures how well samples are clustered.
- Davies–Bouldin Index – Lower value indicates better clustering.
- Calinski–Harabasz Index – Higher value indicates better clustering.
- Adjusted Rand Index (ARI) – Measures similarity between true labels and clusters.

Program:

```
# ============================================================
# UNIFIED ML PERFORMANCE EVALUATION ON WINE DATASET
# ============================================================

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler,label_binarize

# Regression
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# ------------------------------------------------------------
# LOAD DATASET
# ------------------------------------------------------------

wine = load_wine()
X = wine.data
y = wine.target

df = pd.DataFrame(X, columns=wine.feature_names)
df['class'] = y

print("Dataset Shape:", df.shape)
print(df.head())

# ============================================================
# 1️⃣ REGRESSION TASK
# Predict Alcohol using other features
# ============================================================

print("\n======================================")
print("1️⃣ REGRESSION EVALUATION")
print("======================================")

X_reg = df.drop(columns=['alcohol', 'class'])
y_reg = df['alcohol']

X_train, X_test, y_train, y_test = train_test_split(
    X_reg, y_reg, test_size=0.3, random_state=42)

reg_model = LinearRegression()
reg_model.fit(X_train, y_train)

y_pred_reg = reg_model.predict(X_test)
```

```
mae = mean_absolute_error(y_test, y_pred_reg)
mse = mean_squared_error(y_test, y_pred_reg)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred_reg)

print("MAE:", mae)
print("MSE:", mse)
print("RMSE:", rmse)
print("R2 Score:", r2)
```

```
Dataset Shape: (178, 14)
   alcohol  malic_acid   ash  alcalinity_of_ash  magnesium  total_phenols  \
0    14.23        1.71  2.43               15.6      127.0           2.80
1    13.20        1.78  2.14               11.2      100.0           2.65
2    13.16        2.36  2.67               18.6      101.0           2.80
3    14.37        1.95  2.50               16.8      113.0           3.85
4    13.24        2.59  2.87               21.0      118.0           2.80

   flavanoids  nonflavanoid_phenols  proanthocyanins  color_intensity   hue  \
0        3.06                  0.28             2.29             5.64  1.04
1        2.76                  0.26             1.28             4.38  1.05
2        3.24                  0.30             2.81             5.68  1.03
3        3.49                  0.24             2.18             7.80  0.86
4        2.69                  0.39             1.82             4.32  1.04


    od280/od315_of_diluted_wines  proline  class
 0                          3.92   1065.0      0
 1                          3.40   1050.0      0
 2                          3.17   1185.0      0
 3                          3.45   1480.0      0
 4                          2.93    735.0      0

 ======================================
 1️⃣ REGRESSION EVALUATION
 ======================================
 MAE: 0.38604836056924796
 MSE: 0.20175903076082488
 RMSE: 0.44917594632930297
 R2 Score: 0.6241637957918121
```

```
# ==========================================================
# 2️⃣ CLASSIFICATION TASK
# Predict Wine Class + ROC-AUC Curve
# ==========================================================

print("\n================ CLASSIFICATION ================")
# Classification
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (accuracy_score, precision_score,
                             recall_score, f1_score,
                             confusion_matrix,roc_auc_score,
                             roc_curve)

X_clf = df.drop(columns=['class'])
y_clf = df['class']

X_train, X_test, y_train, y_test = train_test_split(
    X_clf, y_clf, test_size=0.3, random_state=42)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

clf_model = LogisticRegression(max_iter=5000)
clf_model.fit(X_train, y_train)

y_pred = clf_model.predict(X_test)
y_prob = clf_model.predict_proba(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision (macro):", precision_score(y_test, y_pred, average='macro'))
print("Recall (macro):", recall_score(y_test, y_pred, average='macro'))
print("F1 Score (macro):", f1_score(y_test, y_pred, average='macro'))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))

# ---------- ROC-AUC (Multi-class: One-vs-Rest) ----------

# Binarize output
y_test_bin = label_binarize(y_test, classes=[0, 1, 2])
```

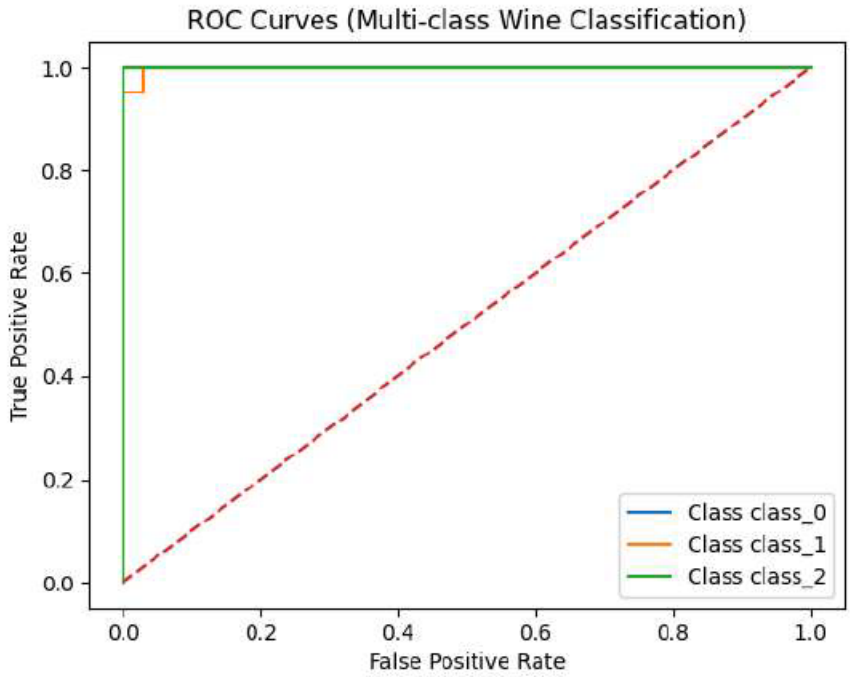
```
roc_auc = roc_auc_score(y_test_bin, y_prob, multi_class='ovr')
print("ROC-AUC Score (OvR):", roc_auc)

# Plot ROC Curves for each class
plt.figure()

for i in range(len(class_names)):
    fpr, tpr, _ = roc_curve(y_test_bin[:, i], y_prob[:, i])
    plt.plot(fpr, tpr, label=f"Class {class_names[i]}")

plt.plot([0, 1], [0, 1], linestyle='--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curves (Multi-class Wine Classification)")
plt.legend()
plt.show()
```

```
================ CLASSIFICATION ================
Accuracy: 0.9814814814814815
Precision (macro): 0.9777777777777779
Recall (macro): 0.9841269841269842
F1 Score (macro): 0.9803756658256239
Confusion Matrix:
 [[19  0  0]
 [ 0 20  1]
 [ 0  0 14]]
ROC-AUC Score (OvR): 0.9995189995189996
```



```
# ==========================================================
# 3 CLUSTERING TASK
# Group Wines Without Labels
# ==========================================================
# Clustering
from sklearn.cluster import KMeans
from sklearn.metrics import (silhouette_score,
                             davies_bouldin_score,
                             calinski_harabasz_score,
                             adjusted_rand_score)
print("\n=====================================")
print("3 CLUSTERING EVALUATION")
print("=====================================")

X_cluster = df.drop(columns=['class'])

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_cluster)

kmeans = KMeans(n_clusters=3, random_state=42)


  y_pred_cluster = kmeans.fit_predict(X_scaled)

  sil = silhouette_score(X_scaled, y_pred_cluster)
  db = davies_bouldin_score(X_scaled, y_pred_cluster)
  ch = calinski_harabasz_score(X_scaled, y_pred_cluster)
  ari = adjusted_rand_score(y, y_pred_cluster)

  print("Silhouette Score:", sil)
  print("Davies-Bouldin Index:", db)
  print("Calinski-Harabasz Index:", ch)
  print("Adjusted Rand Index:", ari)
```

```
=====================================
3 CLUSTERING EVALUATION
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Silhouette Score: 0.2848589191898987
Davies-Bouldin Index: 1.3891879777181646
Calinski-Harabasz Index: 70.9400080031512
Adjusted Rand Index: 0.8974949815093207
```

Conclusion: In this experiment, regression, classification, and clustering models were successfully implemented on the Wine Dataset and evaluated using appropriate performance metrics specific to each learning paradigm. The regression model demonstrated predictive capability through error-based metrics such as MAE, MSE, RMSE, and goodness-of-fit measure R^2 score, highlighting the importance of minimizing prediction error. The classification model effectively distinguished between wine classes, and its performance was comprehensively assessed using Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC–AUC curve, emphasizing both predictive accuracy and class separability. The clustering model identified natural groupings within the dataset, and its quality was validated using internal and external indices such as Silhouette Score, Davies–Bouldin Index, Calinski–Harabasz Index, and Adjusted Rand Index.

Exercise 1: Classification Dataset (Supervised Learning)
Abstract:
Dataset: Iris Flower Dataset
UCI Repository link: https://archive.ics.uci.edu/ml/datasets/iris

The objective of this exercise is to predict the species of an Iris flower based on its sepal and petal measurements. The Iris dataset consists of 150 samples, each described by four numerical features: sepal length, sepal width, petal length, and petal width. Using this dataset, a suitable classification model such as Logistic Regression, K-Nearest Neighbors, or Decision Tree is to be developed. The task involves performing exploratory data analysis and necessary preprocessing, followed by training and validating the model on the given data. The performance of the trained model is to be assessed using standard evaluation metrics including accuracy, precision, recall, F1-score, and confusion matrix, along with cross-validation to ensure reliability. The results obtained from these evaluation strategies are to be interpreted to understand the effectiveness of the model and the significance of each performance metric.

Exercise 2:
Dataset: Wholesale Customers Data Set
UCI Repository link: https://archive.ics.uci.edu/ml/datasets/wholesale+customers
https://www.kaggle.com/datasets/binovi/wholesale-customers-data-set

Abstract: The objective of this exercise is to segment wholesale customers based on their annual spending patterns across different product categories. The Wholesale Customers dataset contains numerical attributes representing yearly expenditure on items such as fresh products, milk, grocery, frozen foods, detergents and paper, and delicatessen. Since no predefined class labels are available, unsupervised learning techniques are to be applied to identify natural groupings within the data. The task involves performing exploratory data analysis and preprocessing steps such as feature scaling, followed by the application of a clustering algorithm like K-Means or Hierarchical Clustering. The optimal number of clusters is to be determined using evaluation techniques such as the Elbow method and Silhouette score. The resulting clusters are to be visualized using dimensionality reduction techniques like PCA, and the cluster characteristics are to be analyzed and interpreted to understand customer segmentation and spending behavior.